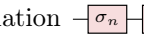
Scope of the Spacetime Bootstrap

(Dated: October 8, 2026)

The spacetime bootstrap establishes a uniform spectral gap from bond reflection positivity and certified finite-scale purity bounds. Bond reflection positivity ensures that the continuous-time spatial transfer operator is self-adjoint, enabling spatial scale doubling dual to imaginary time. We formalize the method as a general criterion for one-dimensional translation-invariant lattice systems and classify standard spin Hamiltonians according to their reflection positivity.

Appendix D: Prerequisites of the Spacetime Bootstrap

The scale-doubling iteration in Ref. [1] comprises three elementary operations:

- *Coupling identities.* Equalities between partition-function ratios on the spacetime torus, such as  . These hold universally for every translation-invariant lattice Hamiltonian.
- *Temporal squaring.* The relation  $\propto \rho_n(2\beta)$ requires only positivity of the Gibbs operator, $e^{-\beta H} = e^{-\beta H/2}(e^{-\beta H/2})^\dagger \geq 0$, which holds generally by Euclidean time reflection.
- *Spatial squaring.* The dual relation  $\propto \sigma_{2n}$ requires positivity of the spatial transfer operator, $\sigma_n = X_\beta^n/Z_n \geq 0$. This holds whenever X_β is self-adjoint ($X_\beta = X_\beta^\dagger$), which additionally guarantees that $Z_n = \sum_i \lambda_i^n$ with real eigenvalues λ_i, permitting ℓ^p norm interpolation to non-dyadic intermediate lengths.

Self-adjointness of X_β represents the spatial counterpart of Euclidean time reflection, and is not automatic. Once it holds, the remaining doubling argument follows directly. The three hypotheses are therefore:

(H1) *Locality.* A translation-invariant chain with finite local dimension and nearest-neighbour terms. Finite-range interactions reduce to this by blocking sites.

(H2) *Bond reflection positivity* (Sec. E).

(H3) *Input.* Both purities are near one at a single scale (Sec. F).

Hypotheses (H1) and (H2) are algebraic properties of the Hamiltonian, whereas (H3) reflects the underlying spectral gap.

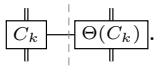
Appendix E: Reflection positivity

Let Θ be an antiunitary map on one site with $\Theta^2 = \pm 1$, for example complex conjugation K in a chosen basis. For an operator C on one site write $\Theta(C) = \Theta C \Theta^{-1}$.

- *Reflection.* Cut the chain between two neighbouring sites. Mirror the chain across the cut and apply Θ on every site. Call the result θ. It is antilinear, it maps the operators on the left half to those on the right half, and $\theta^2 = 1$.
- *Bond form.* A nearest-neighbour Hamiltonian is in *bond reflection-positive form* [5–8] if it reads

$$H = \sum_j D_j - \sum_{j,k} c_k\, C_{k,j}\, \Theta(C_k)_{j+1}, \qquad c_k > 0, \quad \text{(E1)}$$

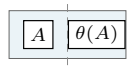
with $\Theta(D) = D$. Each bond event of weight c_k inserts C_k on one site and its mirror image $\Theta(C_k)$ on the next site,

$$-h_{j,j+1} = \sum_k c_k$$ 

The sign in front matters. A term with $c_k < 0$ cannot be absorbed by rescaling C_k, because Θ conjugates the factor i: $(iC) \otimes \Theta(iC) = C \otimes \Theta(C)$.

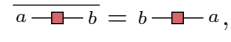
The model of Ref. [1] has this form with $\Theta = K$ in the Cartesian basis and $c_k = 1$. Its generators $G_\alpha = iS^\alpha$ are real, so $-\boldsymbol{S}\cdot\boldsymbol{S} = \sum_\alpha G_\alpha \otimes \overline{G_\alpha}$.

Reflection positivity and transfer matrices. Expand $e^{-\beta H}$ in the bond events that cross the mirror. Each term is a left half followed by its mirror image, so for every operator A on the left half,

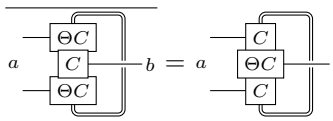
$$\mathrm{Tr}\big[A\,\theta(A)\,e^{-\beta H}\big] =$$  $$= \sum \langle \Phi_A | \Phi_A \rangle \geq 0. \quad \text{(E2)}$$

Reflection positivity began as the Osterwalder–Schrader axiom that rebuilds a Hilbert space and a positive transfer matrix from Euclidean data [2]. On the lattice it gives a self-adjoint, positive transfer matrix [3], and its Cauchy–Schwarz structure drives infrared bounds and chessboard estimates [4, 6, 7]. For spins and Majoranas, Jaffe and Janssens characterize exactly which couplings are reflection positive [5]. The spacetime bootstrap needs only the weaker consequence that follows: the network cut at a single bond is Hermitian. This is the Lüscher statement with space and time exchanged.

Lemma 1 (Mirror is conjugation)**.** *Under Eq. (E1), build the continuous-time MPO as in Ref. [1]. The events are (τ, k), they occur at rate c_k, and $e^{-\tau D}$ propagates between them. Then the transfer kernel obeys*

 *i.e.* $\overline{X_\beta(a,b)} = X_\beta(b,a)$. (E3)

Proof. Open the vertex in imaginary time. Events on the left bond a insert $\Theta(C)$, and events on the right bond b insert C, along a single traced physical loop:

$\overline{X_\beta(a,b)} =$  $= X_\beta(b,a).$

Write $\Theta = VK$. Complex conjugation of a trace conjugates every insertion and keeps the time order. Conjugating the loop by V then maps $\overline{\Theta(C)} \mapsto C$, $\overline{C} \mapsto \Theta(C)$ and $\overline{D} \mapsto D$, because $V\overline{V} = \pm 1$ and $\Theta(D) = D$. The picture that results is the mirror image of the original: the left insertions are now C and the right ones are $\Theta(C)$. □

Theorem 1 (Space behaves like time)**.** *Assume (H1) and (H2). Then X_β is a self-adjoint Hilbert–Schmidt operator on $L^2(\mu_\beta)$. For every even n, $Z_n(\beta) = \sum_i \lambda_i(\beta)^n$ with real λ_i, and $\sigma_n(\beta) \geq 0$. In particular the squaring lemma, the coupling identities and the ℓ^p interpolation of Ref. [1] hold verbatim.*

Proof. The event measure μ_β is positive because $c_k > 0$, and it is finite, with mass $e^{\beta \sum_k c_k}$ per bond. Normalize $\|C_k\| = 1$. The kernel is then bounded, $|X_\beta(a,b)| \leq d\, e^{\beta \|D\|}$, hence Hilbert–Schmidt, and it is Hermitian by Lemma 1. The remaining statements are those of Ref. [1] for a self-adjoint X_β. □

Relation to the sign problem. A Hamiltonian is stoquastic if, in some basis, every configuration of a quantum Monte Carlo expansion has a nonnegative weight. Reflection positivity asks less of individual configurations and more of their structure: a weight may be negative, but each configuration paired with its mirror image contributes a square $|\cdot|^2$.

- In Ref. [1] the event measure is positive, but the entries of W and X_β carry signs, because the G_α are real antisymmetric matrices. A sampler over histories would meet a sign problem; the spectral argument does not, since $\sum_i \lambda_i^n > 0$.
- Neither property implies the other. The quantum Heisenberg ferromagnet is stoquastic but not reflection positive [9]; see also Proposition 1.
- The sign-free fermion Monte Carlo methods based on time reversal or Majorana symmetry exploit the same mirror-squared factorization, $\det M \cdot \overline{\det M}$ [10–13].

Appendix F: The general criterion

Define the one-step loss

$$\delta(\varepsilon) = \tfrac{1}{2}\big[(1-\varepsilon)^{-3} - 1\big]^2. \tag{F1}$$

It lies below the diagonal,  , up to $\varepsilon_* \simeq 0.1273$, the root of $\delta(\varepsilon) = \varepsilon$. Ref. [1] uses the cruder bound $\delta \leq 5\varepsilon^2$, valid for $\varepsilon \leq 1/100$.

Theorem 2 (Spacetime-bootstrap criterion)**.** *Assume (H1) and (H2), and suppose that for some even n_0 and some $\beta_0 > 0$,*

$$P_{\mathrm{s}}(n_0,\beta_0),\ P_\tau(2n_0,\beta_0) \geq 1-\varepsilon_0, \qquad \varepsilon_0 < \varepsilon_*. \tag{F2}$$

Set $\varepsilon_{j+1} = \delta(\varepsilon_j)$, $n_j = 2^j n_0$ and $\beta_j = 2^j \beta_0$. Then for every even $L \geq n_0$ the ground state is unique and

$$\Delta_L \ \geq\ c := \inf_{j\geq 0} \frac{1}{\beta_j} \log \frac{1-3\varepsilon_j}{3\varepsilon_j} \ >\ 0. \tag{F3}$$

Proof. By Theorem 1, the four moves of the doubling estimate in Ref. [1] apply without change. Move 1 gives $P_{\mathrm{s}}(n, 2\beta) \geq (1-\varepsilon)^3$. Move 2 squares in space to get $P_{\mathrm{s}}(2n, 2\beta) \geq 1 - \delta(\varepsilon)$. Move 3 gives $P_\tau(4n, \beta) \geq (1-\varepsilon)^2(1-\delta) \geq (1-\varepsilon)^3$, which requires $\delta \leq \varepsilon$, i.e. $\varepsilon \leq \varepsilon_*$. Move 4 squares in time to get $P_\tau(4n, 2\beta) \geq 1 - \delta(\varepsilon)$. Induction on j gives $P_{\mathrm{s}}(n_j, \beta_j), P_\tau(2n_j, \beta_j) \geq 1 - \varepsilon_j$, with ε_j decreasing because $\delta(\varepsilon) < \varepsilon$ on $(0, \varepsilon_*)$. Since $3\varepsilon_j < 1/2$, the interpolation and gap-recovery steps of Ref. [1] give uniqueness and $e^{-\beta_j \Delta_L} \leq 3\varepsilon_j/(1-3\varepsilon_j)$ whenever $n_j \leq L \leq 2n_j$. Every term of the infimum is positive. Since $\delta(\varepsilon) \leq K\varepsilon^2$ near zero, ε_j falls doubly exponentially. The terms therefore converge to a positive limit, and the infimum is positive. □

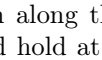
With $(n_0, \beta_0, \varepsilon_0) = (2304, 784, 1/100)$ this is the theorem of Ref. [1], with $c \geq (\log 20)/784$.

Corollary 1 (What (H3) excludes)**.** *Under (H1) and (H2), Eq. (F2) cannot hold at any scale*

(i) if the ground state is k-fold degenerate at the length $2n_0$ with $k \geq 2$, because $P_\tau \leq w_0 \leq 1/k$ there, a spectrum like *;*

(ii) if $\liminf_L \Delta_L = 0$ along even L, because Theorem 2 would contradict it.

Corollary 1 rules out gapless chains, symmetry-broken chains, and chains whose ground state on the ring is asymptotically degenerate.

The converse is expected but not proved. In a gapped chain with a unique ground state and correlation length ξ, one expects $1 - P_{\mathrm{s}} \sim e^{-n/\xi}$ and $1 - P_\tau \sim n\, e^{-\beta\Delta}$. Both vanish along the doubling sequence  , so Eq. (F2) should hold at some finite scale, though perhaps a very large one.

For a conformal critical chain the picture is different. The extensive free energy cancels in each torus ratio, so P_{s} and P_τ approach ratios of the universal torus partition function at a fixed aspect ratio. These are numbers below one, so the defect stalls instead of collapsing.

Appendix G: Classification of Reflection-Positive Hamiltonians

Bilinear spin couplings. Consider spin S with a nearest-neighbour coupling matrix J (real symmetric) and a uniform field $\boldsymbol{h}$,

$$H = \sum_j \boldsymbol{S}_j^{\mathsf{T}} J\, \boldsymbol{S}_{j+1} + \boldsymbol{h}\cdot\boldsymbol{S}_j. \tag{G1}$$

Proposition 1. *Suppose there is a symmetric orthogonal O with $\det O = -1$ such that $-JO \succeq 0$ and $O\boldsymbol{h} = \boldsymbol{h}$. Then Eq. (G1) is in the form of Eq. (E1), for every S. For diagonal $J = \mathrm{diag}(J_x, J_y, J_z)$ with nonzero entries and $\boldsymbol{h} = 0$, such an O exists exactly when an even number of J_x, J_y, J_z are negative (ferromagnetic).*

Proof. $O = -R$ with R the identity or a π rotation. Let T be time reversal, so $TST^{-1} = -S$, and put $\Theta = U_R T$. Then $\Theta\boldsymbol{S}\Theta^{-1} = O\boldsymbol{S}$. Since U_R commutes with T, $\Theta^2 = U_R^2 T^2 = \pm 1$. Factor $-JO = \sum_k v_k v_k^{\mathsf{T}}$ with real v_k and set $C_k = v_k \cdot \boldsymbol{S}$. Then $\sum_k C_k \otimes \Theta(C_k) = \sum \boldsymbol{S}^{\mathsf{T}}(-JO)O\boldsymbol{S} = -\boldsymbol{S}^{\mathsf{T}} J \boldsymbol{S}$, and $\Theta(\boldsymbol{h}\cdot\boldsymbol{S}) = (O\boldsymbol{h})\cdot\boldsymbol{S}$. For diagonal J the polar factor is unique: $O = \mathrm{sign}(-J)$, and $\det O = -1$ counts the ferromagnetic entries modulo two. □

The proposition sorts the standard spin chains as follows.

- *Reflection positive:*
  - the Heisenberg antiferromagnet for every S, with $O = -1$;
  - XXZ chains $S^xS^x + S^yS^y + \Delta S^zS^z$ with $\Delta \geq 0$;
  - the XY model with any signs, since a zero entry frees O_{zz};
  - the transverse-field Ising chain with either sign, with O fixing the field axis.
- *Not of this form:*
  - the Heisenberg ferromagnet (three ferromagnetic entries);
  - XXZ chains with $\Delta < 0$;
  - the antiferromagnet in a uniform field, since $O = -1$ reverses $\boldsymbol{h}$.

The sublattice rotations by π that map antiferromagnets to ferromagnets change two signs at once, so they never alter this parity.

The uniform-field obstruction explains a choice made in Ref. [1], Appendix A. There the boundary chain is built with opposite endpoint fields $\uparrow\cdots\downarrow$, since $\theta(-hS_1^z) = +hS_n^z$ makes that chain mirror symmetric. The equal-field chain $\uparrow\cdots\uparrow$ is reached only through the symmetry U_P.

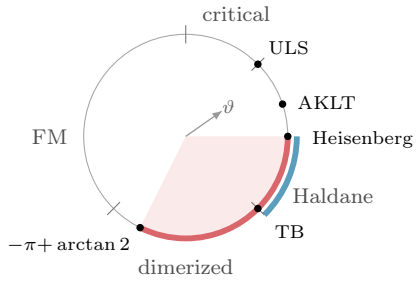
Spin-1 bilinear–biquadratic chain. Consider $H = \sum_j \cos\vartheta\, \boldsymbol{S}_j\cdot\boldsymbol{S}_{j+1} + \sin\vartheta\, (\boldsymbol{S}_j\cdot\boldsymbol{S}_{j+1})^2$. In the Cartesian basis the quadrupoles $Q_{\alpha\beta} = S^\alpha S^\beta$ are real. Their antisymmetric part is $\frac{1}{2}\epsilon_{\alpha\beta\gamma}G_\gamma$, so

$$-h = \left(\cos\vartheta - \tfrac{1}{2}\sin\vartheta\right)\sum_\alpha G_\alpha \otimes \overline{G_\alpha} - \sin\vartheta \sum_{\alpha,\beta} Q^+_{\alpha\beta}\otimes\overline{Q^+_{\alpha\beta}}, \tag{G2}$$

where $Q^+_{\alpha\beta} = \frac{1}{2}(Q_{\alpha\beta} + Q_{\beta\alpha})$; the cross terms cancel between symmetric and antisymmetric parts.

Proposition 2. *With $\Theta = K$ in the Cartesian basis, the bilinear–biquadratic chain is in reflection-positive form if and only if $-\pi + \arctan 2 \leq \vartheta \leq 0$.*

Proof. The condition holds if both coefficients in Eq. (G2) are nonnegative, which is the stated window. For the converse, the form exists exactly when the realigned bond matrix becomes positive semidefinite after some on-site shift. Maximizing its least eigenvalue over all shifts gives a negative value outside the window, for example -0.087 at $\vartheta = -0.66\pi$ and -0.156 at $\vartheta = 0.05\pi$ (Sec. H). □



The red arc in the figure is the reflection-positive window. The blue arc, $-\pi/4 < \vartheta \leq 0$, is where it meets the Haldane phase, so Theorem 2 can apply there. Elsewhere in the window Corollary 1 excludes it: the dimerized ground states are degenerate, and the Takhtajan–Babujian point TB is critical.

The Heisenberg chain sits exactly on the edge of the window. The AKLT and ULS points lie outside it: the positive biquadratic coefficient has the wrong sign. Proving the AKLT gap this way would need a different Θ, a different cut, or blocking: a numerical search over every on-site Θ finds none (Sec. H).

Beyond the chain. Finite-range couplings fit after blocking: put r sites in a unit cell and reflect within the block. Reflection-positive fermion models [10] and two-dimensional lattices with reflection planes [7, 8] supply the structural property (H2). In two dimensions the spatial transfer operator acts on a whole column, so (H3) becomes much harder to verify.

Appendix H: Numerical checks

The checks are short floating-point scripts kept outside this repository. Two tests are used.

- *Gram test* (`rp_check.py`). This tests Eq. (E2) on open chains of 4 and 6 sites, $\beta \in [0.3, 3]$.

The left-half operators A index a Hermitian form, and the test reports its least eigenvalue relative to the largest, minimized over the on-site antiunitary $\Theta = VK$.

- *Realignment test* (`t4.py`). This checks the form of Eq. (E1) directly, with the on-site shift optimized.

The results:

- *Spin-$\frac{1}{2}$ XYZ chain.* All eight sign patterns of $(J_x, J_y, J_z) = (\pm 1, \pm .7, \pm .4)$ follow the parity rule: those with zero or two ferromagnetic entries give $\geq -10^{-12}$ after optimizing V, and the other four give -0.048. The antiferromagnet in a uniform field gives -0.095, and the transverse-field Ising chain gives $\geq -10^{-12}$.

- *Spin-1, Cartesian $\Theta = K$.* The Heisenberg antiferromagnet and $-(\boldsymbol{S}\cdot\boldsymbol{S})^2$ pass. The ferromagnet fails at -0.39 and AKLT fails at -0.04. Over ϑ, the test is nonnegative at the sampled points inside the window of Proposition 2, $\vartheta/\pi \in [-.62, 0]$, and negative at the sampled points outside it, $\vartheta/\pi \in \{-1, -.88, -.75\} \cup [.12, 1]$.

- *Spin-1, every on-site $\Theta = VK$.* Minimizing over $V \in U(3)$ with twelve restarts does not help: AKLT stays at -0.040, ULS at -0.18, $\vartheta = -0.75\pi$ at -0.15, and the ferromagnet improves only to -0.15.

These tests are necessary conditions for a particular θ. A failure excludes that Θ, not every possible proof.

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